Fiscal Year:	FY 2012	**Task Last Updated:**	FY 05/14/2012
PI Name:	Kiss, John Z Ph.D.		
Project Title:	Fractional gravity studies of sensory mechanisms in root phototropism–a reflight experiment (TROPI-2)		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	jzk@fit.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	336-334-5241
Organization Name:	University of North Carolina-Greensboro		
PI Address 1:	College of Arts & Sciences, 105 Foust Building		
PI Address 2:	PO Box 26170		
PI Web Page:			
City:	Greensboro	**State:**	NC
Zip Code:	27402-6170	**Congressional District:**	13
Comments:	NOTE: PI moved to Florida Institute of Technology from University of North Carolina-Greensboro in May 2024 (Ed., 2/6/25). NOTE: PI moved to University of North Carolina-Greensboro from University of Mississippi in July 2016 (Ed., 6/13/17) NOTE: PI moved to U Miss from Miami (Ohio) University in September 2012 (Ed., 8/10/2012)		
Project Type:	Flight	**Solicitation / Funding Source:**	NOT AVAILABLE
Start Date:	04/01/2009	**End Date:**	03/31/2012
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	3
No. of Bachelor's Candidates:	5	**Monitoring Center:**	NASA ARC
Contact Monitor:	Sun, Sidney C	**Contact Phone:**	650-604-4835
Contact Email:	Sidney.C.Sun@nasa.gov		
Flight Program:	Shuttle/ISS		
Flight Assignment:	ISS STS-130 NOTE: New end date is 3/31/2012 per NSSC info (Ed., 2/19/2011)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Edelmann, Richard (Miami University) Correll, Melanie (University of Florida)		
Grant/Contract No.:	NNX09AF11G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The major goals of this research are (1) to better understand cellular mechanisms of phototropism in plants and (2) to determine the effects and influence of gravity on light perception in plants. Specifically, we plan to use microgravity to facilitate studies of phototropism in roots and hypocotyls of young seedlings of the plant Arabidopsis thaliana. Results of our ground-based studies suggest that there are two photosensory systems that elicit phototropic responses in roots: a previously identified blue-light photoreceptor system and a red light phytochrome-based mechanism that we have recently discovered. Because of the interfering effect of the strong gravitropic response in roots, microgravity conditions are needed to effectively study this novel red-light-dependent phototropic response. Intermediate or fractional gravity studies (less than the nominal 1g on Earth) were planned to further study the interaction between tropisms. TROPI-1 was successfully performed on the ISS during late 2006. We obtained data on seedlings grown in microgravity and from the 1g control. However, one problem encountered during TROPI-1 was low seed germination due to long storage periods (8 months) in flight hardware. Thus, the originally proposed fractional gravity studies were not performed. TROPI-2 provides an opportunity to regain the results from these important fractional gravity experiments. These experiments will provide a better understanding of how plants integrate sensory input from multiple light and gravity perception systems. This information is important for growing plants on long-term space missions as part of life support systems. The fractional gravity studies contain 0.16g (Moon) and 0.38g (Mars) treatments, so information to be obtained is relevant to the exploration agenda at NASA. We used the European Modular Cultivation System (EMCS) on the ISS during Increment 22 in February/March 2010.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	From a basic science perspective, we will gain new insight into cellular/molecular events in light sensing. In the long term, through greater understanding of how plants perceive and respond to light, it will be possible to improve crop productivity. It also should be possible to modify agricultural crop plants with genetically improved phytochrome or their downstream signaling molecules to develop transgenic crop species with more fine tuned light-responsive developmental traits to better fit their ambient light environments.
Task Progress:	While there is a great deal of knowledge regarding plant growth and development in microgravity aboard orbiting spacecraft, there is little information available about these parameters in reduced or fractional gravity conditions (less than the nominal 1g on Earth). Thus, in these experiments using the European Modular Cultivation System on the International Space Station, we studied the interaction between phototropism and gravitropism in the WT and mutants of phytochrome A and B of Arabidopis thaliana. Fractional gravity and the 1g control were provided by centrifuges in the spaceflight hardware, and unidirectional red and blue illumination followed a white light growth period in the time line of the space experiments. The existence of red-light-based positive phototropism in hypocotyls of seedlings that is mediated by phytochrome was confirmed in these microgravity experiments. Fractional gravity studies showed an attenuation of red-light-based phototropism in both roots and hypocotyls of seedlings occurring due to gravitational accelerations ranging from 0.lg to 0.3g. In contrast, blue-light negative phototropism in roots, which was enhanced in microgravity compared to the 1g control, showed a significant attenuation at 0.3g. In addition, our studies suggest that the well known red-light enhancement of blue-light-induced phototropism in hypocotyls is likely due to an indirect effect by the attenuation of gravitropism. However, red-light enhancement of root blue-light-based phototropism may occur via a more direct effect on the phototropism system itself, most likely through the phytochrome photoreceptors. To our knowledge, these experiments represent the first to examine the behavior of flowering plants in fractional or reduced gravity conditions.
Bibliography Type:	Description: (Last Updated: 11/25/2025)
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Abstracts for Journals and Proceedings	Kiss JZ, Millar KDL, Edelmann RE. "Phototropism and nutation of seedlings grown in microgravity." 27th Annual Meeting of the American Society for Gravitational and Space Biology, San Jose, CA, November 3-6, 2011. Program and Abstracts. 27th Annual Meeting of the American Society for Gravitational and Space Biology, San Jose, CA, November 3-6, 2011. p. 103. , Nov-2011
Articles in Peer-reviewed Journals	Millar KD, Johnson CM, Edelmann RE, Kiss JZ. "An endogenous growth pattern of roots is revealed in seedlings grown in microgravity." Astrobiology. 2011 Oct;11(8):787-97. Epub 2011 Oct 4. PMID: 21970704 ; http://dx.doi.org/10.1089/ast.2011.0699 , Oct-2011
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Articles in Peer-reviewed Journals	Hopkins JA, Kiss JZ. "Phototropism and gravitropism in transgenic lines of Arabidopsis altered in the phytochrome pathway." Physiol Plant. 2012 Mar 1. [Epub ahead of print] PMID: 22380624 ; http://dx.doi.org/10.1111/j.1399-3054.2012.01607.x , Mar-2012
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Journal/Magazine covers	Millar KD, Johnson CM, Edelmann RE, Kiss JZ. "Cover in Astrobiology 2011 issue for article, 'An endogenous growth pattern of roots is revealed in seedlings grown in microgravity.'" Astrobiology. 2011 Oct;11(8):787-97. PubMed PMID: 21970704 ; http://www.liebertonline.com/action/showLargeCover?issue=40220197 , Oct-2011